

$\Rightarrow$ Given Differentiable,

$$\Rightarrow f'(0^-) = f'(0^+),$$

$$f'(0^-) = \frac{d}{dx} (\sin 2x)_{(0)}$$

$$= 2 \cos(2x)_0 = 2$$

$$f'(0^+) = b \Rightarrow 2$$

$\Rightarrow$ continuous too.

$$\Rightarrow f(0^+) = f(0^-).$$

$$\Rightarrow \sin(2 \times 0) = a + 2x \Rightarrow a = 0.$$

$$\Rightarrow a = 0, \quad b = 2, \quad \checkmark.$$